B.G.D | Hardware Issue11 electrical E1

Functional connectivity / Q75 + R1.1 / not a fabrication or power release

Module pin layout is logical. Named terminals do not assign physical connector cavities.

10-1 Pack and distribution

File: 10-1.kicad_sch

10-2 Pack and distribution

File: 10-2.kicad_sch

20-1 Regulated rails

File: 20-1.kicad_sch

20-2 Regulated rails

File: 20-2.kicad_sch

30-1 FC / radio / sensors

File: 30-1.kicad_sch

30-2 FC / radio / sensors

File: 30-2.kicad_sch

40-1 Compute / switches

File: 40-1.kicad_sch

40-2 Compute / switches

File: 40-2.kicad_sch

40-3 Compute / switches

File: 40-3.kicad_sch

50-1 PWM / lights / release

File: 50-1.kicad_sch

50-2 PWM / lights / release

File: 50-2.kicad_sch

50-3 PWM / lights / release

File: 50-3.kicad_sch

50-4 PWM / lights / release

File: 50-4.kicad_sch

60-1 Charge / cable holds

File: 60-1.kicad_sch

Sheet: /
File: bgd-master.kicad_sch

Title:

Size: A3	Date:	**Rev:**
KiCad E.D.A. 10.0.6		Id: 1/15

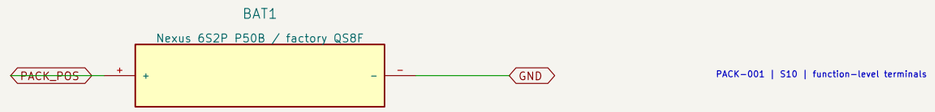
10.1 Pack, monitored distribution and motor phases

DRAFT LOGICAL CIRCUIT — no physical pad or connector view is implied.

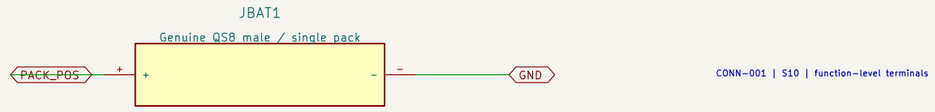
PACK_POS -> INA shunt -> BUS_POS: all load branches remain downstream.

Common GND uses dedicated sized returns; carbon is never a conductor.

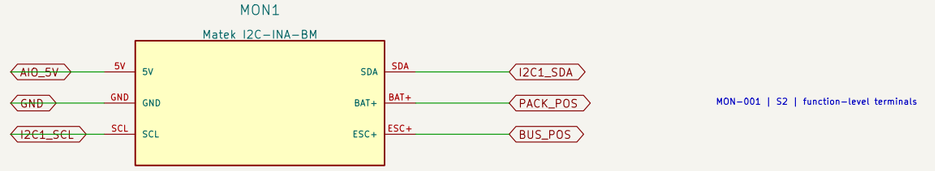
Motor A/B/C are commissioning IDs; actual pad identity/rotation remains held.



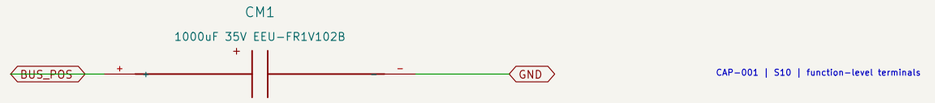
PACK-001 | S10 | function-level terminals



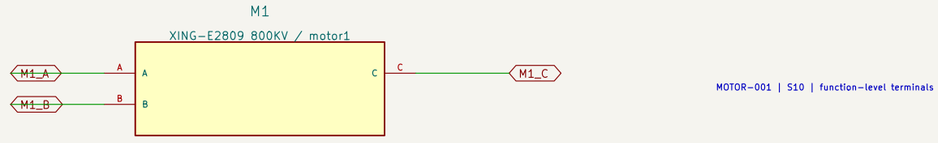
CONN-001 | S10 | function-level terminals



MON-001 | S2 | function-level terminals



CAP-001 | S10 | function-level terminals



MOTOR-001 | S10 | function-level terminals

NO FABRICATION / POWER RELEASE

Sheet: /10-1 Pack and distribution/
File: 10-1.kicad_sch

Title: BGD electrical / group 10

Size: A3 | Date: 2026-09-05 | **Rev: E1 LOGICAL DRAFT**
KiCad E.D.A. 10.0.6 | Id: 2/15

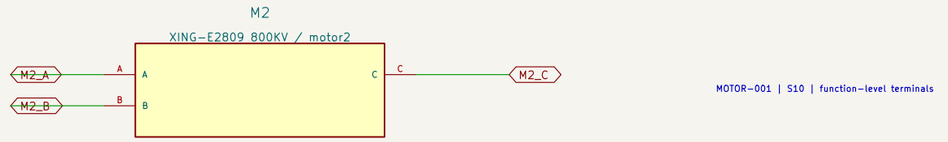
10.2 Pack, monitored distribution and motor phases

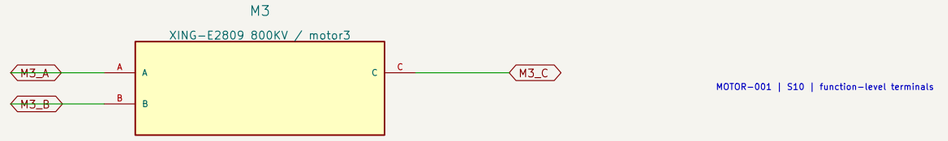
DRAFT LOGICAL CIRCUIT — no physical pad or connector view is implied.

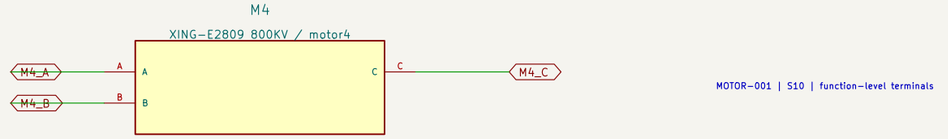
PACK_POS -> INA shunt -> BUS_POS: all load branches remain downstream.

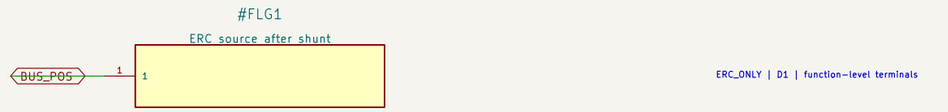
Common GND uses dedicated sized returns; carbon is never a conductor.

Motor A/B/C are commissioning IDs; actual pad identity/rotation remains held.









NO FABRICATION / POWER RELEASE

Sheet: /10-2 Pack and distribution/
File: 10-2.kicad_sch

Title: BGD electrical / group 10

Size: A3	Date: 2026-09-05	**Rev: E1 LOGICAL DRAFT**
KiCad E.D.A. 10.0.6		Id: 3/15

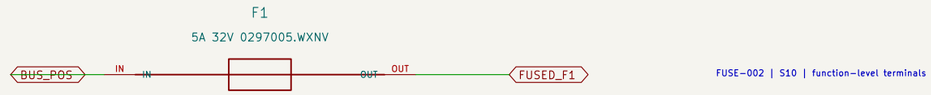
20.1 Regulated rails and branch protection

Regulator outputs stay separate. Shared GND does not permit sharing thin returns.

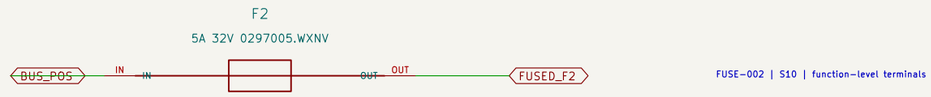
F1/F2:5A32V INPUT fuses; not coordinated output–load or main–battery protection.

DFR1015: remove5V bridge; select3.3V only; supply from AIO12V.

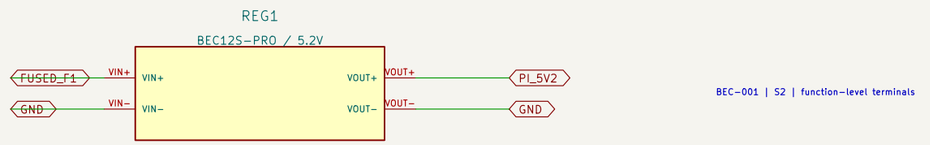
IR load and servo peak demand remain provisional; no completed power–budget pass.



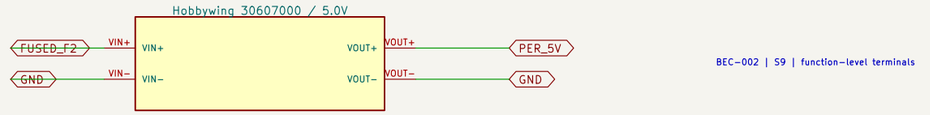
FUSE–002 | S10 | function–level terminals



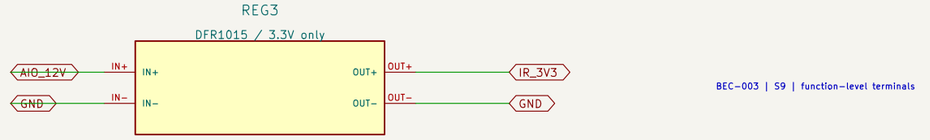
FUSE–002 | S10 | function–level terminals



BEC–001 | S2 | function–level terminals



BEC–002 | S9 | function–level terminals



BEC–003 | S9 | function–level terminals

NO FABRICATION / POWER RELEASE

Sheet: /20–1 Regulated rails/
File: 20–1.kicad_sch

Title: BGD electrical / group 20

Size: A3	Date: 2026–09–05	**Rev: E1 LOGICAL DRAFT**
KiCad E.D.A. 10.0.6		Id: 4/15

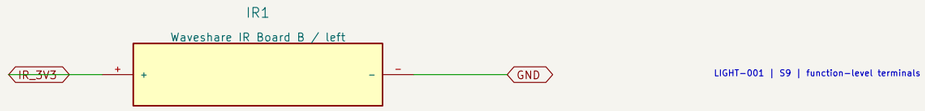
20.2 Regulated rails and branch protection

Regulator outputs stay separate. Shared GND does not permit sharing thin returns.

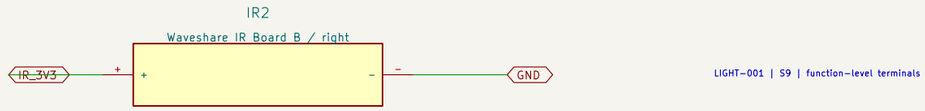
F1/F2:5A32V INPUT fuses; not coordinated output-load or main-battery protection.

DFR1015: remove5V bridge; select3.3V only; supply from AIO12V.

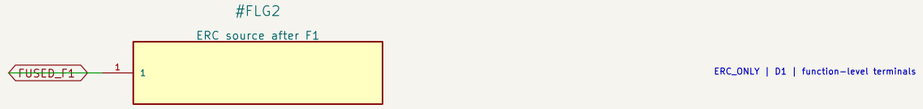
IR load and servo peak demand remain provisional; no completed power-budget pass.



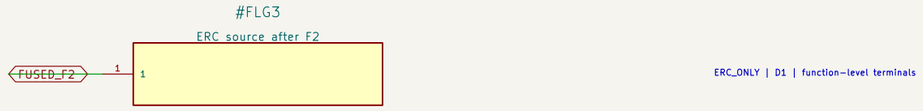
LIGHT-001 | S9 | function-level terminals



LIGHT-001 | S9 | function-level terminals



ERC_ONLY | D1 | function-level terminals



ERC_ONLY | D1 | function-level terminals

NO FABRICATION / POWER RELEASE

Sheet: /20-2 Regulated rails/
File: 20-2.kicad_sch

Title: BGD electrical / group 20

Size: A3	Date: 2026-09-05	**Rev: E1 LOGICAL DRAFT**
KiCad E.D.A. 10.0.6		Id: 5/15

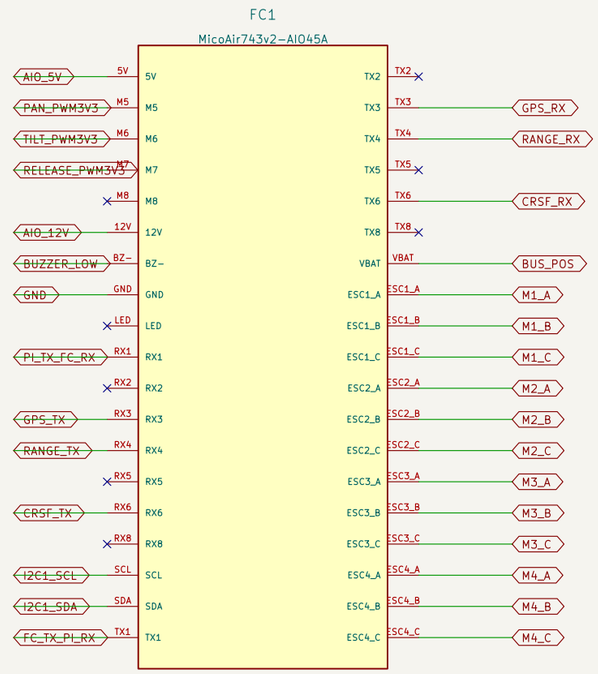
30.1 Flight controller, radio and sensors

FC symbol uses named manufacturer pads; layout is NOT the physical board view.

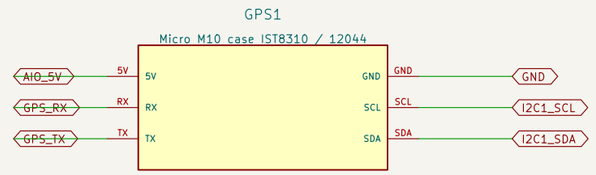
I2C pull-ups/revisions/rise time remain held; do not add blind pull-ups.

Selected GPS is IST8310 SKU12044. Reserved UARTs/M8/LED remain unused.

TFmini supply must stay4.9-5.1V at its connector; UART is3.3V.



FC-001 | S1 | function-level terminals



GNSS-001 | S6 | function-level terminals

NO FABRICATION / POWER RELEASE

Sheet: /30-1 FC / radio / sensors/
File: 30-1.kicad_sch

Title: BGD electrical / group 30

Size: A3 | Date: 2026-09-05 | **Rev: E1 LOGICAL DRAFT**
KiCad E.D.A. 10.0.6 | Id: 6/15

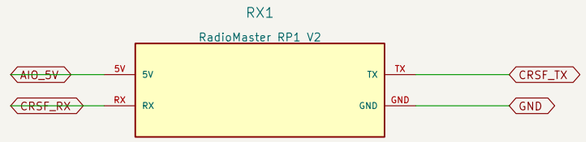
30.2 Flight controller, radio and sensors

FC symbol uses named manufacturer pads; layout is NOT the physical board view.

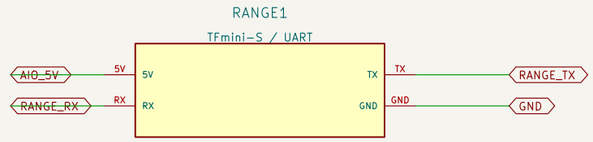
I2C pull-ups/revisions/rise time remain held; do not add blind pull-ups.

Selected GPS is IST8310 SKU12044. Reserved UARTs/M8/LED remain unused.

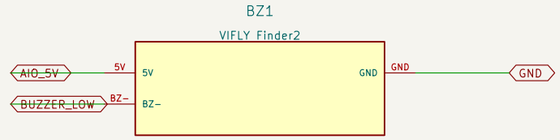
TFmini supply must stay4.9-5.1V at its connector; UART is3.3V.



RX-001 | S8 | function-level terminals



SENSE-002 | S7 | function-level terminals



BUZZ-001 | S8 | function-level terminals

NO FABRICATION / POWER RELEASE

Sheet: /30-2 FC / radio / sensors/
File: 30-2.kicad_sch

Title: BGD electrical / group 30

Size: A3 | Date: 2026-09-05 | **Rev: E1 LOGICAL DRAFT**
KiCad E.D.A. 10.0.6 | Id: 7/15

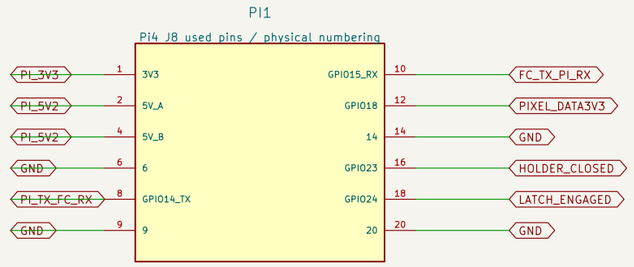
40.1 Compute power, UART and holder sensing

PI1 numbers are PHYSICAL J8 pins; used subset only. Carrier2310 mates1:1.

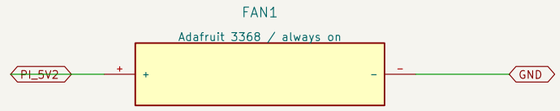
Equal22AWG tails to2/4 and6/9; inspect header joints/current sharing.

No simultaneous Pi USB-C power. GPIO18 PWM audio/other waveform owners disabled.

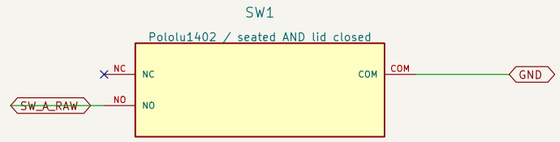
Switches: aircraft mounted, NO-to-GND only when safe; short-to-GND is not detected.



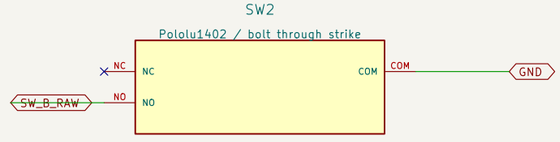
PI-001 | S3 | function-level terminals



COOL-001 | S3 | function-level terminals



ACT-003 | S5 | function-level terminals



ACT-004 | S5 | function-level terminals

NO FABRICATION / POWER RELEASE

Sheet: /40-1 Compute / switches/
File: 40-1.kicad_sch

Title: BGD electrical / group 40

Size: A3	Date: 2026-09-05	**Rev: E1 LOGICAL DRAFT**
KiCad E.D.A. 10.0.6		Id: 8/15

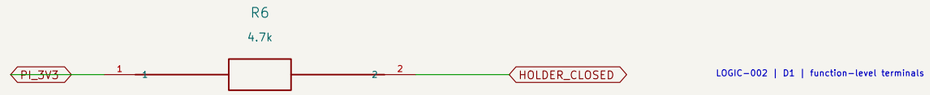
40.2 Compute power, UART and holder sensing

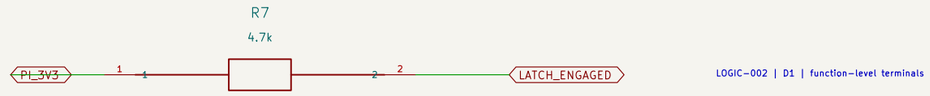
PI1 numbers are PHYSICAL J8 pins; used subset only. Carrier2310 mates1:1.

Equal22AWG tails to2/4 and6/9; inspect header joints/current sharing.

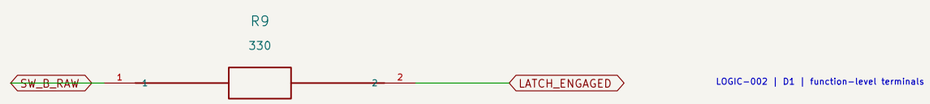
No simultaneous Pi USB-C power. GPIO18 PWM audio/other waveform owners disabled.

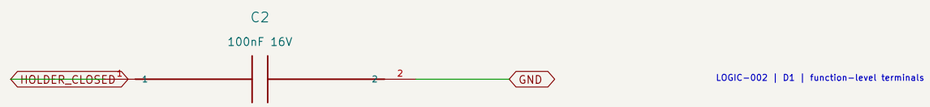
Switches: aircraft mounted, NO-to-GND only when safe; short-to-GND is not detected.











NO FABRICATION / POWER RELEASE

Sheet: /40-2 Compute / switches/
File: 40-2.kicad_sch

Title: BGD electrical / group 40

Size: A3	Date: 2026-09-05	**Rev: E1 LOGICAL DRAFT**
KiCad E.D.A. 10.0.6		Id: 9/15

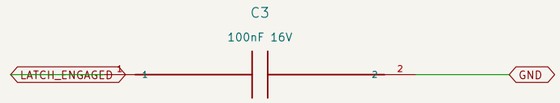
40.3 Compute power, UART and holder sensing

PI1 numbers are PHYSICAL J8 pins; used subset only. Carrier2310 mates1:1.

Equal22AWG tails to2/4 and6/9; inspect header joints/current sharing.

No simultaneous Pi USB-C power. GPIO18 PWM audio/other waveform owners disabled.

Switches: aircraft mounted, NO-to-GND only when safe; short-to-GND is not detected.



LOGIC-002 | D1 | function-level terminals

NO FABRICATION / POWER RELEASE

Sheet: /40-3 Compute / switches/
File: 40-3.kicad_sch

Title: BGD electrical / group 40

Size: A3	Date: 2026-09-05	**Rev: E1 LOGICAL DRAFT**
KiCad E.D.A. 10.0.6		Id: 10/15

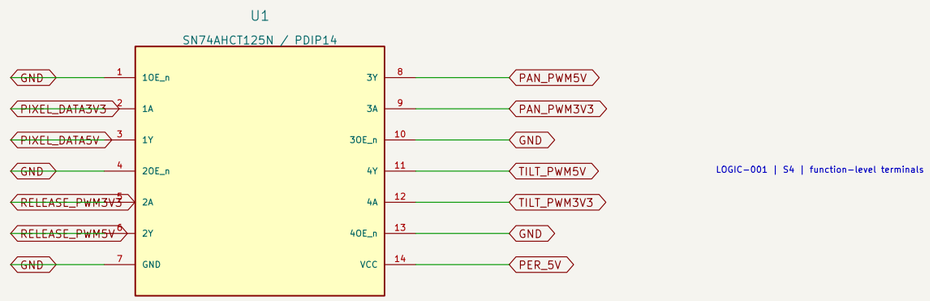
50.1 Translated PWM, lighting and cargo release

U1 peripheral5V island must never bridge the carrier Pi5V copper.

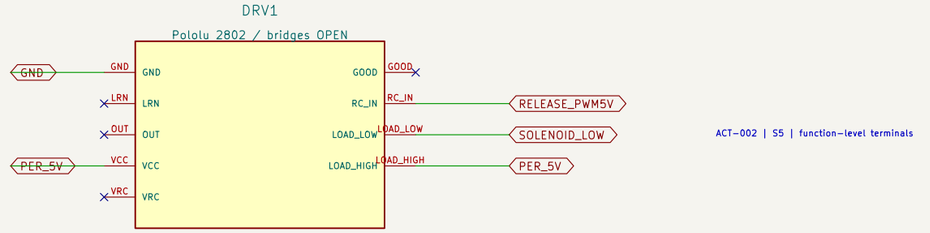
All OEs LOW; all inputs10k pulldown. R5 at receiving LED end.

DRV1 power bridges OPEN; LOAD_LOW is switched, never a permanent ground.

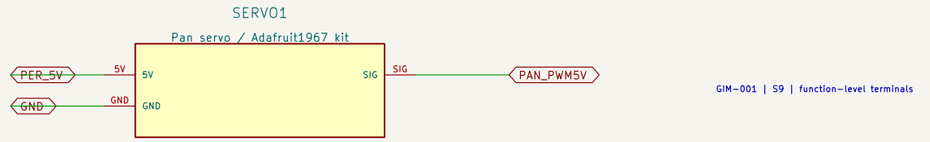
FC-owned release timeout is mandatory: continuing stale PWM defeats pulse-loss safe-start.



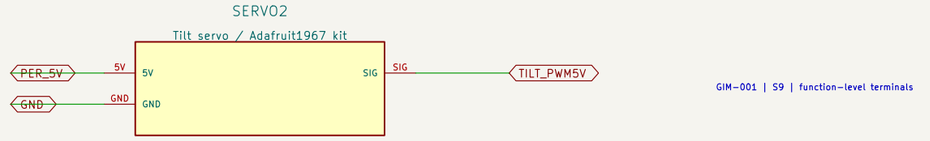
LOGIC-001 | S4 | function-level terminals



ACT-002 | S5 | function-level terminals



GIM-001 | S9 | function-level terminals



GIM-001 | S9 | function-level terminals

NO FABRICATION / POWER RELEASE

Sheet: /50-1 PWM / lights / release/
File: 50-1.kicad_sch

Title: BGD electrical / group 50

Size: A3	Date: 2026-09-05	**Rev: E1 LOGICAL DRAFT**
KiCad E.D.A. 10.0.6		Id: 11/15

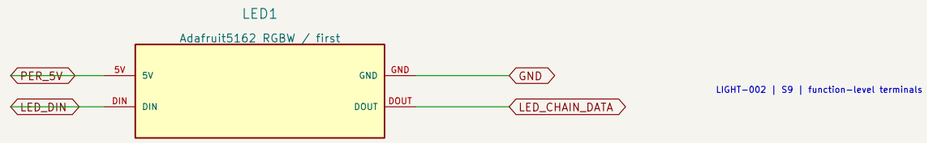
50.2 Translated PWM, lighting and cargo release

U1 peripheral5V island must never bridge the carrier Pi5V copper.

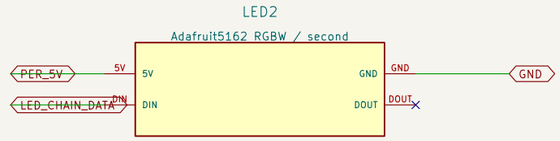
All OEs LOW; all inputs10k pulldown. R5 at receiving LED end.

DRV1 power bridges OPEN; LOAD_LOW is switched, never a permanent ground.

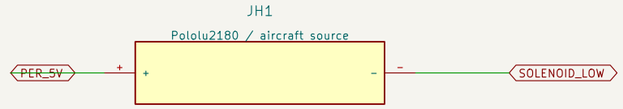
FC-owned release timeout is mandatory: continuing stale PWM defeats pulse-loss safe-start.



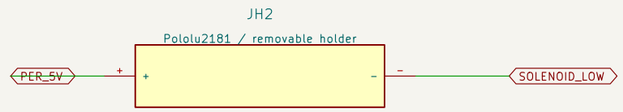
LIGHT-002 | S9 | function-level terminals



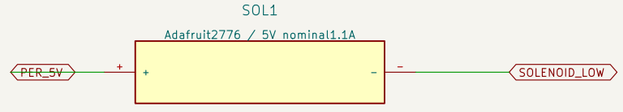
LIGHT-002 | S9 | function-level terminals



ACT-005 | S11 | function-level terminals



ACT-005 | S11 | function-level terminals



ACT-001 | S5 | function-level terminals

NO FABRICATION / POWER RELEASE

Sheet: /50-2 PWM / lights / release/
File: 50-2.kicad_sch

Title: BGD electrical / group 50

Size: A3	Date: 2026-09-05	**Rev: E1 LOGICAL DRAFT**
KiCad E.D.A. 10.0.6		Id: 12/15

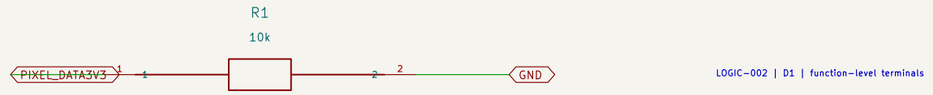
50.3 Translated PWM, lighting and cargo release

U1 peripheral5V island must never bridge the carrier Pi5V copper.

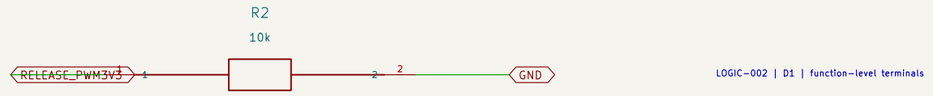
All OEs LOW; all inputs10k pulldown. R5 at receiving LED end.

DRV1 power bridges OPEN; LOAD_LOW is switched, never a permanent ground.

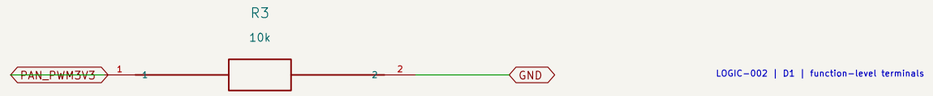
FC-owned release timeout is mandatory: continuing stale PWM defeats pulse-loss safe-start.



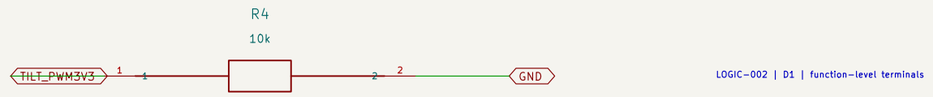
LOGIC-002 | D1 | function-level terminals



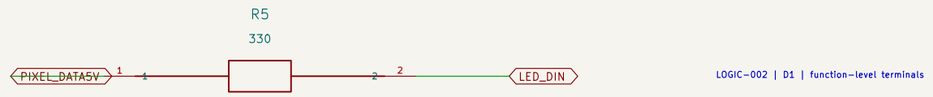
LOGIC-002 | D1 | function-level terminals



LOGIC-002 | D1 | function-level terminals



LOGIC-002 | D1 | function-level terminals



LOGIC-002 | D1 | function-level terminals

NO FABRICATION / POWER RELEASE

Sheet: /50-3 PWM / lights / release/
File: 50-3.kicad_sch

Title: BGD electrical / group 50

Size: A3	Date: 2026-09-05	**Rev: E1 LOGICAL DRAFT**
KiCad E.D.A. 10.0.6		Id: 13/15

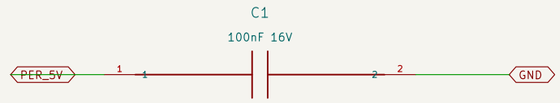
50.4 Translated PWM, lighting and cargo release

U1 peripheral5V island must never bridge the carrier Pi5V copper.

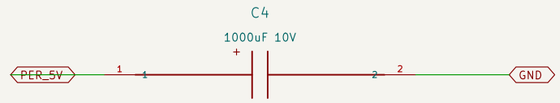
All OEs LOW: all inputs10k pulldown. R5 at receiving LED end.

DRV1 power bridges OPEN; LOAD_LOW is switched, never a permanent ground.

FC-owned release timeout is mandatory: continuing stale PWM defeats pulse-loss safe-start.



LOGIC-002 | D1 | function-level terminals



LOGIC-002 | D1 | function-level terminals

NO FABRICATION / POWER RELEASE

Sheet: /50-4 PWM / lights / release/
File: 50-4.kicad_sch

Title: BGD electrical / group 50

Size: A3	Date: 2026-09-05	**Rev: E1 LOGICAL DRAFT**
KiCad E.D.A. 10.0.6		Id: 14/15

60.1 Ground charging and factory-cable boundaries

CHARGE ONLY /5A MAX. Charge net names are deliberately separate from flight nets.

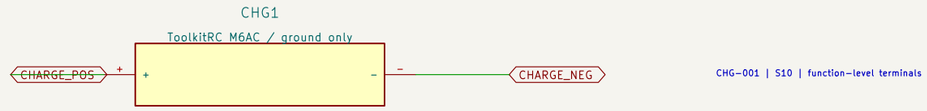
No connection to a live flight configuration is represented or authorized.

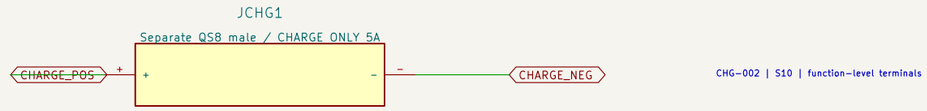
W100 CSI15: retain factory cable; orientation/motion loop requires mechanical review.

W101 USB500mm + W102 D500100mm: cable internals/inrush/route remain held.

W103 RF: exact factory mate. W106 balance: verify all7contacts B0..B6 before use.

NC123-134: power bridges open, spare ports isolated; see reconciliation register.





NO FABRICATION / POWER RELEASE

Sheet: /60-1 Charge / cable holds/
File: 60-1.kicad_sch

Title: BGD electrical / group 60

Size: A3	Date: 2026-09-05	**Rev: E1 LOGICAL DRAFT**
KiCad E.D.A. 10.0.6		Id: 15/15